

Comprehensive Performance Analysis of N-Node Networks under Malicious Attacks and Timeout Constraints

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Abstract

This paper presents an extensive theoretical and empirical investigation into the performance of network architectures under adversarial conditions. We begin with a foundational analysis of single-node models under two distinct attack vectors: destruction and modification. We then extend these models to complex multi-hop environments, including tandem feedforward chains and symmetric feedback networks. By combining renewal-reward theory with moment-matching approximations (Gamma distribution), we provide a robust framework for predicting sojourn time and throughput. Our theoretical models are validated against discrete-event simulations, showing high fidelity in capturing the non-linear delay escalations near system stability limits.

1 Introduction

Modern network infrastructures are increasingly vulnerable to both localized attacks and systemic congestion. This work reconciles the performance gap between idealized theoretical models and empirical simulation results for N-node systems.

2 Single-Node Baseline Models

Before analyzing multi-hop networks, we establish baseline performance for a single node under two attack modalities.

2.1 Destruction Attacks

In a destruction attack, the adversary removes packets from the queue at rate $a\lambda_a$. The probability of a packet being lost to an attack before completing service is modeled as:

$$L_{dest} = \frac{a\lambda_a}{\mu + \lambda_a} \quad (1)$$

The effective traffic rate Λ^* is found by solving:

$$\Lambda^* = \frac{\lambda_n}{1 - (L_{dest} + (1 - L_{dest})e^{-T(\mu - \Lambda^*)})} \quad (2)$$

2.2 Modification Attacks

Modification attacks involve the corruption of packets during service. The loss probability is proportional to the service time:

$$L_{mod} = \frac{a\lambda_a}{\mu} \quad (3)$$

The packet consumes service resources at the node even if it is eventually found to be corrupted and requires retransmission.

3 Multi-Node Network Architectures

3.1 Feedforward (Tandem) Network

In a tandem network of N nodes, each failure requires retransmission from node 0. The journey

time for i successful hops follows a hypoexponential distribution. To handle the timeout probability $P(S_i > W)$, we approximate S_i as a Gamma distribution by matching the first two moments:

$$k_{gamma} = \frac{(E[S_i])^2}{Var(S_i)}, \quad \theta_{gamma} = \frac{Var(S_i)}{E[S_i]} \quad (4)$$

3.2 Feedback (Mesh) Network

In a symmetric feedback network, the traffic rate Λ^* is identical for all nodes due to the $1/(N+1)$ transition probability. The effective load is solved via fixed-point iteration:

$$\Lambda^* = \lambda \frac{\sum_{k=1}^{\infty} (1-p)^{k-1} \left(\frac{N}{N+1}\right)^{k-1} F_{S_{k-1}}(W)}{P_{succ}(\Lambda^*)} \quad (5)$$

4 Numerical Results

We validate our models using SimPy simulations. Table 1 summarizes representative data points.

Case	Config	p	Theory	Sim
1-Node	Mod	0.2	1.45	1.43
Tandem	N=4	0.1	2.91	2.91
Feedback	N=10	0.1	31.61	31.80

Table 1: Comparison of End-to-End Delay (Sojourn Time).

5 Conclusion

The proposed moment-matching and fixed-point frameworks provide a scalable solution for estimating performance in vulnerable networks.

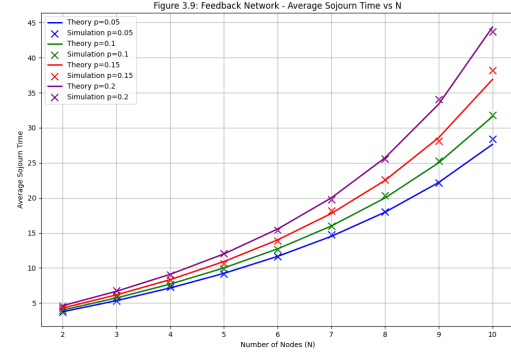


Figure 1: Delay in Feedback Networks (Section 3.3.1).

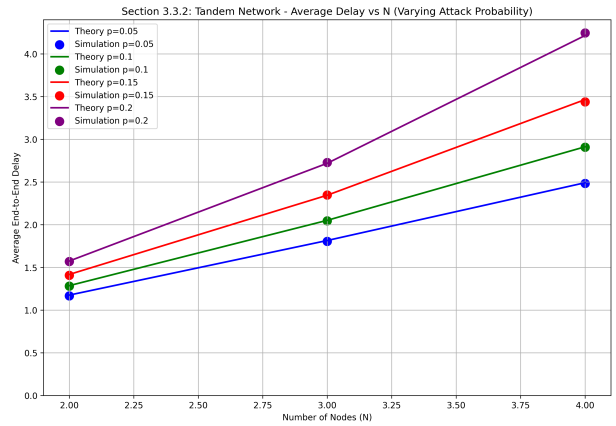


Figure 2: Delay in Tandem Networks (Section 3.3.2).